

AICT_ASSIGNMENT

2026-S-CYS-12

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SECTION A

Problem 1:

Pseudo Code of theater seating chart:

```
START
Create 2D array seats containing ticket prices
FUNCTION displayChart
    FOR each row in seats
        Display row
    END FOR
END FUNCTION
REPEAT
    Call displayChart
    Display:
        1. Select Seat
        2. Select by Price
        3. Exit
    Input choice
    IF choice = 1 THEN
        Input row number
        Input seat number
        IF seat is available (not 0) THEN
            Set seat value to 0
            Display "Seat booked successfully"
        ELSE
            Display "Seat already sold"
        END IF
    ELSE IF choice = 2 THEN
        Input desired price
        Search all rows and seats
        IF seat with matching price is found THEN
            Set seat value to 0
        END IF
    END IF
END REPEAT
```

```

        Display seat location
    ELSE
        Display "No available seat at that price"
    END IF
ELSE IF choice = 3 THEN
    Display "Thank you"
    EXIT loop
ELSE
    Display "Invalid choice"
END IF
UNTIL choice = 3
STOP

```

Code of theaeter seating chart:

```

seats = [
    [10,10,10,10,10,10,10,10,10,10],
    [10,10,10,10,10,10,10,10,10,10],
    [10,10,10,10,10,10,10,10,10,10],
    [10,10,20,20,20,20,20,20,10,10],
    [10,10,20,20,20,20,20,20,10,10],
    [10,10,20,20,20,20,20,20,10,10],
    [20,20,30,30,40,40,30,30,20,20],
    [20,30,30,40,50,50,40,30,30,20],
    [30,40,50,50,50,50,50,40,30,30],
    [30,40,50,50,50,50,50,40,30,30]]
def display_chart():
    print("\nSeat Numbers")
    print("1 2 3 4 5 6 7 8 9 10")
    for i in range(len(seats)):
        print(f"Row {i+1:2}", end=" ")
        for price in seats[i]:
            print(f"{price:2}", end=" ")
        print()
while True:
    display_chart()
    choice = input(
        "\nChoose an option:\n"
        "1. Select a seat\n"
        "2. Select by price\n"
        "3. Exit\n"
        "Enter choice: "
    )
    if choice == "1":
        row = int(input("Enter row (1-10): ")) - 1
        col = int(input("Enter seat number (1-10): ")) - 1
        if 0 <= row < 10 and 0 <= col < 10:
            if seats[row][col] != 0:
                print(f"Seat booked. Price = ${seats[row][col]}")
                seats[row][col] = 0
            else:
                print("Seat already sold.")
        else:

```

```

        print("Invalid seat.")
    elif choice == "2":
        price = int(input("Enter desired price: "))
        found = False
        for r in range(10):
            for c in range(10):
                if seats[r][c] == price:
                    print(f"Seat found at Row {r+1}, Seat {c+1}")
                    seats[r][c] = 0
                    found = True
                    break
            if found:
                break
        if not found:
            print("No available seat at that price.")
    elif choice == "3":
        print("Thank you!")
        break
    else:
        print("Invalid choice.")

```

Pseudo Code for best customer:

```

START
FUNCTION Best_Customer(sales, customers)
    maxSales ← first value in sales
    bestCustomer ← first value in customers
    FOR each remaining sale in sales
        IF current sale > maxSales THEN
            maxSales ← current sale
            bestCustomer ← corresponding customer
        END IF
    END FOR
    RETURN bestCustomer
END FUNCTION
Create empty list sales
Create empty list customers
REPEAT
    Input purchaseAmount
    IF purchaseAmount = 0 THEN
        EXIT loop
    END IF
    Input customerName
    Add purchaseAmount to sales
    Add customerName to customers
UNTIL purchaseAmount = 0
IF sales list is not empty THEN
    best ← Best_Customer(sales, customers)

```

```
        Display "Best Customer of the Day: ", best
ELSE
    Display "No customer data entered."
END IF
STOP
```

Code for best customer:

```
def Best_Customer(sales, customers):
    max_sales = sales[0]
    best_customer = customers[0]
    for i in range(1, len(sales)):
        if sales[i] > max_sales:
            max_sales = sales[i]
            best_customer = customers[i]
    return best_customer

sales = []
customers = []
while True:
    amount = float(input("Enter purchase amount (Enter 0 to stop): "))
    if amount == 0:
        break
    name = input("Enter customer name: ")
    sales.append(amount)
    customers.append(name)
if len(sales) > 0:
    best = Best_Customer(sales, customers)
    print("\nBest Customer of the Day:", best)
else:
    print("No customer data entered.")
```